



Name: _____

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Learning Objectives

- Evaluate a function using function notation $f(x)$
- Determine the domain and range of a function
- Compute slope and write the equation of a line
- Find the composition and inverse of a function

Simplify each expression completely. Show all steps and circle your final answer.

Function composition

1. Let $f(x) = 1x + -5$ and $g(x) = 3x + -2$. Find $(f \circ g)(-1) = f(g(-1))$.

$$f(x) = 1x + -5, g(x) = 3x + -2, x = -1$$

Answer: _____

2. Let $f(x) = 3x + 1$ and $g(x) = 4x + -1$. Find $(f \circ g)(-2) = f(g(-2))$.

$$f(x) = 3x + 1, g(x) = 4x + -1, x = -2$$

Answer: _____

3. Let $f(x) = 5x + 0$ and $g(x) = 2x + 3$. Find $(f \circ g)(3) = f(g(3))$.

$$f(x) = 5x + 0, g(x) = 2x + 3, x = 3$$

Answer: _____

4. Let $f(x) = 2x + 3$ and $g(x) = 5x + -3$. Find $(f \circ g)(-1) = f(g(-1))$.

$$f(x) = 2x + 3, g(x) = 5x + -3, x = -1$$

Answer: _____

5. Let $f(x) = 2x + 2$ and $g(x) = 5x + -1$. Find $(f \circ g)(1) = f(g(1))$.

$$f(x) = 2x + 2, g(x) = 5x + -1, x = 1$$

Answer: _____

6. Let $f(x) = 1x + 0$ and $g(x) = 5x + 4$. Find $(f \circ g)(-2) = f(g(-2))$.

$$f(x) = 1x + 0, g(x) = 5x + 4, x = -2$$

Answer: _____



7. Let $f(x) = 3x + -1$ and $g(x) = 5x + -3$. Find $(f \circ g)(0) = f(g(0))$.

$$f(x) = 3x + -1, \quad g(x) = 5x + -3, \quad x = 0$$

Answer: _____

8. Let $f(x) = 4x + 2$ and $g(x) = 3x + -1$. Find $(f \circ g)(1) = f(g(1))$.

$$f(x) = 4x + 2, \quad g(x) = 3x + -1, \quad x = 1$$

Answer: _____

9. Let $f(x) = 1x + -1$ and $g(x) = 4x + -2$. Find $(f \circ g)(4) = f(g(4))$.

$$f(x) = 1x + -1, \quad g(x) = 4x + -2, \quad x = 4$$

Answer: _____

10. Let $f(x) = 1x + -4$ and $g(x) = 2x + -4$. Find $(f \circ g)(2) = f(g(2))$.

$$f(x) = 1x + -4, \quad g(x) = 2x + -4, \quad x = 2$$

Answer: _____

Evaluating functions

11. Let $f(x) = 1x + 1$. Find $f(-2)$.

$$f(x) = 1x + 1, \quad x = -2$$

Answer: _____

12. Let $f(x) = 1x + 4$. Find $f(-4)$.

$$f(x) = 1x + 4, \quad x = -4$$

Answer: _____

13. Let $f(x) = 1x + 3$. Find $f(2)$.

$$f(x) = 1x + 3, \quad x = 2$$

Answer: _____

14. Let $f(x) = 3x + -7$. Find $f(4)$.

$$f(x) = 3x - 7, \quad x = 4$$

Answer: _____



15. Let $f(x) = 1x + -2$. Find $f(-2)$.

$$f(x) = 1x - 2, \quad x = -2$$

Answer: _____

16. Let $f(x) = 5x + -2$. Find $f(-1)$.

$$f(x) = 5x - 2, \quad x = -1$$

Answer: _____

17. Let $f(x) = 4x + -3$. Find $f(4)$.

$$f(x) = 4x - 3, \quad x = 4$$

Answer: _____

18. Let $f(x) = 4x + -5$. Find $f(6)$.

$$f(x) = 4x - 5, \quad x = 6$$

Answer: _____

19. Let $f(x) = 1x + -5$. Find $f(2)$.

$$f(x) = 1x - 5, \quad x = 2$$

Answer: _____

20. Let $f(x) = 1x + 1$. Find $f(3)$.

$$f(x) = 1x + 1, \quad x = 3$$

Answer: _____

Slope and linear equations

21. Find the slope of the line through (1, 3) and (6, 7).

$$(1, 3), (6, 7)$$

Answer: _____

22. Find the slope of the line through (0, -4) and (4, -3).

$$(0, -4), (4, -3)$$

Answer: _____

23. Find the slope of the line through (2, -1) and (5, 2).

$$(2, -1), (5, 2)$$

Answer: _____



24. Find the slope of the line through $(-2, 2)$ and $(3, 6)$.

$(-2, 2), (3, 6)$

Answer: _____

25. Find the slope of the line through $(-5, 2)$ and $(1, 3)$.

$(-5, 2), (1, 3)$

Answer: _____

26. Find the slope of the line through $(-2, -4)$ and $(4, -4)$.

$(-2, -4), (4, -4)$

Answer: _____

27. Find the slope of the line through $(-3, -4)$ and $(4, -8)$.

$(-3, -4), (4, -8)$

Answer: _____

28. Find the slope of the line through $(1, -1)$ and $(4, 0)$.

$(1, -1), (4, 0)$

Answer: _____

29. Find the slope of the line through $(2, 3)$ and $(3, 5)$.

$(2, 3), (3, 5)$

Answer: _____

30. Find the slope of the line through $(0, 3)$ and $(5, 8)$.

$(0, 3), (5, 8)$

Answer: _____





Topics: Function composition, Evaluating functions, Slope and linear equations. All answers verified by independent computation.

Solutions



Function composition

1. Let $f(x) = 1x + -5$ and $g(x) = 3x + -2$. Find $(f \circ g)(-1) = f(g(-1))$.

$$f(x) = 1x + -5, g(x) = 3x + -2, x = -1$$

$$\rightarrow g(-1) = 3(-1) + -2 = -5.$$

$$\rightarrow f(-5) = 1(-5) + -5 = -10.$$

Answer: $(f \circ g)(-1) = f(-5) = -10$

2. Let $f(x) = 3x + 1$ and $g(x) = 4x + -1$. Find $(f \circ g)(-2) = f(g(-2))$.

$$f(x) = 3x + 1, g(x) = 4x + -1, x = -2$$

$$\rightarrow g(-2) = 4(-2) + -1 = -9.$$

$$\rightarrow f(-9) = 3(-9) + 1 = -26.$$

Answer: $(f \circ g)(-2) = f(-9) = -26$

3. Let $f(x) = 5x + 0$ and $g(x) = 2x + 3$. Find $(f \circ g)(3) = f(g(3))$.

$$f(x) = 5x + 0, g(x) = 2x + 3, x = 3$$

$$\rightarrow g(3) = 2(3) + 3 = 9.$$

$$\rightarrow f(9) = 5(9) + 0 = 45.$$

Answer: $(f \circ g)(3) = f(9) = 45$

4. Let $f(x) = 2x + 3$ and $g(x) = 5x + -3$. Find $(f \circ g)(-1) = f(g(-1))$.

$$f(x) = 2x + 3, g(x) = 5x + -3, x = -1$$

$$\rightarrow g(-1) = 5(-1) + -3 = -8.$$

$$\rightarrow f(-8) = 2(-8) + 3 = -13.$$

Answer: $(f \circ g)(-1) = f(-8) = -13$

5. Let $f(x) = 2x + 2$ and $g(x) = 5x + -1$. Find $(f \circ g)(1) = f(g(1))$.

$$f(x) = 2x + 2, g(x) = 5x + -1, x = 1$$

$$\rightarrow g(1) = 5(1) + -1 = 4.$$

$$\rightarrow f(4) = 2(4) + 2 = 10.$$

Answer: $(f \circ g)(1) = f(4) = 10$

6. Let $f(x) = 1x + 0$ and $g(x) = 5x + 4$. Find $(f \circ g)(-2) = f(g(-2))$.

$$f(x) = 1x + 0, g(x) = 5x + 4, x = -2$$

$$\rightarrow g(-2) = 5(-2) + 4 = -6.$$

$$\rightarrow f(-6) = 1(-6) + 0 = -6.$$

Answer: $(f \circ g)(-2) = f(-6) = -6$



7. Let $f(x) = 3x + -1$ and $g(x) = 5x + -3$. Find $(f \circ g)(0) = f(g(0))$.

$$f(x) = 3x + -1, g(x) = 5x + -3, x = 0$$

$$\rightarrow g(0) = 5(0) + -3 = -3.$$

$$\rightarrow f(-3) = 3(-3) + -1 = -10.$$

Answer: $(f \circ g)(0) = f(-3) = -10$

8. Let $f(x) = 4x + 2$ and $g(x) = 3x + -1$. Find $(f \circ g)(1) = f(g(1))$.

$$f(x) = 4x + 2, g(x) = 3x + -1, x = 1$$

$$\rightarrow g(1) = 3(1) + -1 = 2.$$

$$\rightarrow f(2) = 4(2) + 2 = 10.$$

Answer: $(f \circ g)(1) = f(2) = 10$

9. Let $f(x) = 1x + -1$ and $g(x) = 4x + -2$. Find $(f \circ g)(4) = f(g(4))$.

$$f(x) = 1x + -1, g(x) = 4x + -2, x = 4$$

$$\rightarrow g(4) = 4(4) + -2 = 14.$$

$$\rightarrow f(14) = 1(14) + -1 = 13.$$

Answer: $(f \circ g)(4) = f(14) = 13$

10. Let $f(x) = 1x + -4$ and $g(x) = 2x + -4$. Find $(f \circ g)(2) = f(g(2))$.

$$f(x) = 1x + -4, g(x) = 2x + -4, x = 2$$

$$\rightarrow g(2) = 2(2) + -4 = 0.$$

$$\rightarrow f(0) = 1(0) + -4 = -4.$$

Answer: $(f \circ g)(2) = f(0) = -4$



Evaluating functions

11. Let $f(x) = 1x + 1$. Find $f(-2)$.

$$f(x) = 1x + 1, \quad x = -2$$

$$\rightarrow f(-2) = 1(-2) + 1 = -2 + 1 = -1.$$

Answer: $f(-2) = -1$

12. Let $f(x) = 1x + 4$. Find $f(-4)$.

$$f(x) = 1x + 4, \quad x = -4$$

$$\rightarrow f(-4) = 1(-4) + 4 = -4 + 4 = 0.$$

Answer: $f(-4) = 0$

13. Let $f(x) = 1x + 3$. Find $f(2)$.

$$f(x) = 1x + 3, \quad x = 2$$

$$\rightarrow f(2) = 1(2) + 3 = 2 + 3 = 5.$$

Answer: $f(2) = 5$

14. Let $f(x) = 3x + -7$. Find $f(4)$.

$$f(x) = 3x - 7, \quad x = 4$$

$$\rightarrow f(4) = 3(4) + -7 = 12 + -7 = 5.$$

Answer: $f(4) = 5$

15. Let $f(x) = 1x + -2$. Find $f(-2)$.

$$f(x) = 1x - 2, \quad x = -2$$

$$\rightarrow f(-2) = 1(-2) + -2 = -2 + -2 = -4.$$

Answer: $f(-2) = -4$

16. Let $f(x) = 5x + -2$. Find $f(-1)$.

$$f(x) = 5x - 2, \quad x = -1$$

$$\rightarrow f(-1) = 5(-1) + -2 = -5 + -2 = -7.$$

Answer: $f(-1) = -7$

17. Let $f(x) = 4x + -3$. Find $f(4)$.

$$f(x) = 4x - 3, \quad x = 4$$

$$\rightarrow f(4) = 4(4) + -3 = 16 + -3 = 13.$$

Answer: $f(4) = 13$



18. Let $f(x) = 4x + -5$. Find $f(6)$.

$$f(x) = 4x - 5, \quad x = 6$$

$$\rightarrow f(6) = 4(6) + -5 = 24 + -5 = 19.$$

Answer: $f(6) = 19$

19. Let $f(x) = 1x + -5$. Find $f(2)$.

$$f(x) = 1x - 5, \quad x = 2$$

$$\rightarrow f(2) = 1(2) + -5 = 2 + -5 = -3.$$

Answer: $f(2) = -3$

20. Let $f(x) = 1x + 1$. Find $f(3)$.

$$f(x) = 1x + 1, \quad x = 3$$

$$\rightarrow f(3) = 1(3) + 1 = 3 + 1 = 4.$$

Answer: $f(3) = 4$



Slope and linear equations

21. Find the slope of the line through (1, 3) and (6, 7).

(1, 3), (6, 7)

$$\rightarrow m = (y_2 - y_1) / (x_2 - x_1) = 4/5 = 4/5.$$

Answer: $m = \frac{\Delta y}{\Delta x} = \frac{4}{5} = 4/5$

22. Find the slope of the line through (0, -4) and (4, -3).

(0, -4), (4, -3)

$$\rightarrow m = (y_2 - y_1) / (x_2 - x_1) = 1/4 = 1/4.$$

Answer: $m = \frac{\Delta y}{\Delta x} = \frac{1}{4} = 1/4$

23. Find the slope of the line through (2, -1) and (5, 2).

(2, -1), (5, 2)

$$\rightarrow m = (y_2 - y_1) / (x_2 - x_1) = 3/3 = 1.$$

Answer: $m = \frac{\Delta y}{\Delta x} = \frac{3}{3} = 1$

24. Find the slope of the line through (-2, 2) and (3, 6).

(-2, 2), (3, 6)

$$\rightarrow m = (y_2 - y_1) / (x_2 - x_1) = 4/5 = 4/5.$$

Answer: $m = \frac{\Delta y}{\Delta x} = \frac{4}{5} = 4/5$

25. Find the slope of the line through (-5, 2) and (1, 3).

(-5, 2), (1, 3)

$$\rightarrow m = (y_2 - y_1) / (x_2 - x_1) = 1/6 = 1/6.$$

Answer: $m = \frac{\Delta y}{\Delta x} = \frac{1}{6} = 1/6$

26. Find the slope of the line through (-2, -4) and (4, -4).

(-2, -4), (4, -4)

$$\rightarrow m = (y_2 - y_1) / (x_2 - x_1) = 0/6 = 0.$$

Answer: $m = \frac{\Delta y}{\Delta x} = \frac{0}{6} = 0$



27. Find the slope of the line through (-3, -4) and (4, -8).

$$(-3, -4), (4, -8)$$

$$\rightarrow m = (y_2 - y_1) / (x_2 - x_1) = -4/7 = -4/7.$$

Answer: $m = \frac{\Delta y}{\Delta x} = \frac{-4}{7} = -4/7$

28. Find the slope of the line through (1, -1) and (4, 0).

$$(1, -1), (4, 0)$$

$$\rightarrow m = (y_2 - y_1) / (x_2 - x_1) = 1/3 = 1/3.$$

Answer: $m = \frac{\Delta y}{\Delta x} = \frac{1}{3} = 1/3$

29. Find the slope of the line through (2, 3) and (3, 5).

$$(2, 3), (3, 5)$$

$$\rightarrow m = (y_2 - y_1) / (x_2 - x_1) = 2/1 = 2.$$

Answer: $m = \frac{\Delta y}{\Delta x} = \frac{2}{1} = 2$

30. Find the slope of the line through (0, 3) and (5, 8).

$$(0, 3), (5, 8)$$

$$\rightarrow m = (y_2 - y_1) / (x_2 - x_1) = 5/5 = 1.$$

Answer: $m = \frac{\Delta y}{\Delta x} = \frac{5}{5} = 1$

